

第十五课预习报告

invert4.c

- 作用: 演示位运算在整数的低位反转中的应用, 通过异或掩码实现指定数量比特位的翻转。
- 功能: 用户输入整数, 程序显示对应的二进制, 然后反转低4位并展示改变后的结果。
- 演示效果:

```
1  /* invert4.c -- using bit operations to display binary */
2  #include <stdio.h>
3  #include <limits.h>
4  char * itobs(int, char *);
5  void show_bstr(const char *);
6  int invert_end(int num, int bits);
7
8  int main(void)
9  {
10     char bin_str[CHAR_BIT * sizeof(int) + 1];
11
12     int number;
13
14     puts("Enter integers and see them in binary.");
15     puts("Non-numeric input terminates program.");
16     while (scanf("%d", &number) == 1)
17     {
18         itobs(number, bin_str);
19         printf("%d is\n", number);
20         show_bstr(bin_str);
21         putchar('\n');
22         number = invert_end(number, 4);
23         printf("Inverting the last 4 bits gives\n");
24         show_bstr(itobs(number, bin_str));
25         putchar('\n');
26     }
27     puts("Bye!");
28
29     return 0;
30 }
31
```

```

32 char * itobs(int n, char * ps)
33 {
34     int i;
35     const static int size = CHAR_BIT * sizeof(int);
36
37     for (i = size - 1; i ≥ 0; i--, n ≫= 1)
38         ps[i] = (01 & n) + '0';
39     ps[size] = '\\0';
40
41     return ps;
42 }
43
44 /* show binary string in blocks of 4 */
45 void show_bstr(const char * str)
46 {
47     int i = 0;
48
49     while (str[i]) /* not the null character */
50     {
51         putchar(str[i]);
52         if(++i % 4 == 0 && str[i])
53             putchar(' ');
54     }
55 }
56
57 int invert_end(int num, int bits)
58 {
59     int mask = 0;
60     int bitval = 1;
61
62     while (bits-- > 0)
63     {
64         mask |= bitval;
65         bitval ≪= 1;
66     }
67
68     return num ^ mask;
69 }

```

```

1 gcc invert4.c -o invert4 && echo "12`n13`nabc" | ./invert4
2 Enter integers and see them in binary.
3 Non-numeric input terminates program.
4 12 is
5 0000 0000 0000 0000 0000 0000 0000 1100
6 Inverting the last 4 bits gives
7 0000 0000 0000 0000 0000 0000 0000 0011
8 13 is
9 0000 0000 0000 0000 0000 0000 0000 1101
10 Inverting the last 4 bits gives
11 0000 0000 0000 0000 0000 0000 0000 0010
12 Bye!

```

fields.c

- 作用: 演示位域在结构体中的定义与使用, 展示字段压缩存储的效果。
- 功能: 定义包含位域的结构体, 显示其大小与字段值, 说明位域如何节省空间。
- 演示效果:

```

1  /* fields.c -- defining and using bit fields */
2  #include <stdio.h>
3  #include <stdbool.h>
4
5  struct {
6      unsigned int autofil : 1;
7      unsigned int bps : 4;
8      unsigned int ccs : 4;
9      unsigned int : 4;
10     bool status : 1;
11 } printer;
12
13 int main(void)
14 {
15     printf("sizeof printer = %zu\n", sizeof(printer));
16     printer.autofil = 1;
17     printer.bps = 9;
18     printer.ccs = 12;
19     printer.status = true;
20     printf("autofil = %u\n", printer.autofil);
21     printf("bps = %u\n", printer.bps);
22     printf("ccs = %u\n", printer.ccs);

```

```

23     printf("status = %d\n", printer.status);
24     return 0;
25 }

```

```

1  gcc fields.c -o fields && echo "abc" | ./fields
2  sizeof printer = 4
3  autofil = 1
4  bps = 9
5  ccs = 12
6  status = 1

```

dualview.c

- 作用: 演示位域与位运算结合, 通过联合体实现同一数据的不同视图访问。
- 功能: 用户输入整数, 程序显示其二进制表示, 并通过位域结构访问各部分位值。
- 演示效果:

```

1  /* dualview.c -- bit fields and bitwise operators */
2  #include <stdio.h>
3  #include <limits.h> // for CHAR_BIT
4  char * itobs(int, char *);
5  void show_bstr(const char *);
6
7  struct views {
8      unsigned int : 6;
9      unsigned int field1 : 3;
10     unsigned int field2 : 3;
11     unsigned int field3 : 3;
12     unsigned int : 1;
13 };
14
15 union {
16     struct views bf;
17     int value;
18 } view;
19
20 int main(void)
21 {
22     char bin_str[CHAR_BIT * sizeof(int) + 1];
23     int number;
24

```

```

25     puts("Enter integers and see them in binary.");
26     puts("Non-numeric input terminates program.");
27     while (scanf("%d", &number) == 1)
28     {
29         itobs(number, bin_str);
30         printf("%d is ", number);
31         show_bstr(bin_str);
32         putchar('\n');
33         view.value = number;
34         printf("field1 = %d, field2 = %d, field3 = %d\n",
35             view.bf.field1, view.bf.field2, view.bf.field3);
36     }
37     puts("Bye!");
38
39     return 0;
40 }
41
42 char * itobs(int n, char * ps)
43 {
44     int i;
45     const static int size = CHAR_BIT * sizeof(int);
46
47     for (i = size - 1; i ≥ 0; i--, n >>= 1)
48         ps[i] = (01 & n) + '0';
49     ps[size] = '\0';
50
51     return ps;
52 }
53
54 void show_bstr(const char * str)
55 {
56     int i = 0;
57
58     while (str[i])
59     {
60         putchar(str[i]);
61         if (++i % 4 == 0 && str[i])
62             putchar(' ');
63     }
64 }

```

```

1 gcc dualview.c -o dualview && echo "15\nabc" | ./dualview
2 Enter integers and see them in binary.
3 Non-numeric input terminates program.
4 15 is 0000 0000 0000 0000 0000 0000 0000 1111
5 field1 = 7, field2 = 1, field3 = 0
6 Bye!

```

align.c

- 作用: 演示 C11 中 `_Alignof` 与 `_Alignas` 的使用方式, 说明类型和变量在内存中的对齐要求。
- 功能: 输出不同类型的对齐要求, 并显示各变量的内存地址。
- 演示效果:

```

1 // align.c -- using _Alignof and _Alignas (C11)
2
3 #include <stdio.h>
4 int main(void)
5 {
6     double dx;
7     char ca;
8     char cx;
9     double dz;
10    char cb;
11    char _Alignas(double) cz;
12
13    printf("char alignment:  %zd\n", _Alignof(char));
14    printf("double alignment: %zd\n", _Alignof(double));
15    printf("&dx: %p\n", &dx);
16    printf("&ca: %p\n", &ca);
17    printf("&cx: %p\n", &cx);
18    printf("&dz: %p\n", &dz);
19    printf("&cb: %p\n", &cb);
20    printf("&cz: %p\n", &cz);
21
22    return 0;
23 }

```

```

1 gcc align.c -o align && ./align
2 char alignment: 1
3 double alignment: 8
4 &dx: 00000070b8dffdb8
5 &ca: 00000070b8dffdb7
6 &cx: 00000070b8dffdb6
7 &dz: 00000070b8dffda8
8 &cb: 00000070b8dffda7
9 &cz: 00000070b8dffda0

```

binbit.c

- 作用: 演示位运算将整数转换为二进制字符串并分组显示。
- 功能: 用户输入整数，程序将其转换为二进制形式，并按每4位分组输出。
- 演示效果:

```

1  /* binbit.c -- using bit operations to display binary */
2  #include <stdio.h>
3  #include <limits.h> // for CHAR_BIT, # of bits per char
4  char * itobs(int, char *);
5  void show_bstr(const char *);
6
7  int main(void)
8  {
9      char bin_str[CHAR_BIT * sizeof(int) + 1];
10     int number;
11
12     puts("Enter integers and see them in binary.");
13     puts("Non-numeric input terminates program.");
14     while (scanf("%d", &number) == 1)
15     {
16         itobs(number, bin_str);
17         printf("%d is ", number);
18         show_bstr(bin_str);
19         putchar('\n');
20     }
21     puts("Bye!");
22
23     return 0;
24 }
25

```

```

26 char * itobs(int n, char * ps)
27 {
28     int i;
29     const static int size = CHAR_BIT * sizeof(int);
30
31     for (i = size - 1; i ≥ 0; i--, n >>= 1)
32         ps[i] = (01 & n) + '0';
33     ps[size] = '\0';
34
35     return ps;
36 }
37
38 /* show binary string in blocks of 4 */
39 void show_bstr(const char * str)
40 {
41     int i = 0;
42
43     while (str[i]) /* not the null character */
44     {
45         putchar(str[i]);
46         if(++i % 4 == 0 && str[i])
47             putchar(' ');
48     }
49 }

```

```

1 gcc binbit.c -o binbit && echo "5\n255\nabc" | ./binbit
2 Enter integers and see them in binary.
3 Non-numeric input terminates program.
4 5 is 0000 0000 0000 0000 0000 0000 0000 0101
5 255 is 0000 0000 0000 0000 0000 0000 1111 1111
6 Bye!

```